

Robot terrain traversability planning using only a camera and vision foundation models

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It is one of the ideas mentioned in the canvas. Firstly, to evaluate our method, we plan to get videos in different environments, for example, as shown in the image below, where we expect our method to choose a path using the ramp instead of the stairs. Similar environment videos will be taken inside, where the floor will be cluttered with small objects such as empty cans, water bottles, boxes etc. These objects can be categorized into those that are safe to step on or push aside (eg. empty can) and those that are not safe to step on (eg. a box). In this way we can assign a traversability score based on whether the robot can safely move past them. We also plan to include cases with hanging cloth, which may be considered traversable and observe how traversability score will change over time as the robot's view of the cloth evolves from far to close-up images.



Secondly, we want to use SAM to segment the image into multiple areas, and these masks will be numbered. (Similar to [1]). Then these segments will be fed to either a VLLM like ChatGPT or VLMs like CLIP and ask it to assign traversability values ranging from 0 to 1. Once we have identified a sufficiently well-traversable region, we can estimate the slope of this region either by finding the gradient of the depthmap from monocular depth estimators like DepthAnything or by trying to fit a plane by sampling points from the traversable region. A scoring function can be designed that assigns a high score for high slope regions (slope score)

Thirdly, when we have a traversability score from a VLLM and a slope score from depth, we can fuse these two pieces of information to create a costmap in the image space. With the help of this image space, we can plan a path using A*.

How to maintain a traversability distribution over time for consistency is still not decided and needs further research.

Components to tackle - in the context of achieving relevant grades

We are aiming to target grade **A**.

Grade E Deliverables :

We are planning to set up depthAnything and SAM as individual modules at first.

DepthAnything : Setup depthAnything and run it on our environments we hope to test the final pipeline on. Analyse the data (only limited to each frame for now) and come up with a sufficient terrain scoring function.

SAM : Setup SAM on the same images, run the segmentation and process the results to obtain the contours of unique entities in a way that makes intuitive sense when we feed that information along with the RGB image to the VLLM. (Does not make sense to send all the different possible submasks generated by SAM)

Traversability score (initial analysis) : Give the RGB image and the contour information to the VLLM and ask it to assign traversability scores to each of the unique entities segmented by SAM. Analyse the results and check if results are sensible.

Path planning : Assume that we have “perfect” information about traversability information. Provide start and goal positions in the pixel space, run A* and compute the path (all on just one image!) Check if it makes sense.

Grade C deliverables

We extend the grade E pipeline to evaluate it on a video feed.

SAM + VLLM pipeline : Analyse the consistency of the VLLM based traversability score assignment on a video feed of the environment it is expected to traverse. Implement traversability scores as a temporal probability distribution, taking into account the scores assigned to the segment across video feeds.

Come up with a pipeline to “track” the segments across image frames so that we make sure we update the distribution correctly for the same “object” in the image.

depthAnything: Analyse the consistency of depthAnything across video feeds and finetune the terrain scoring function to be robust to these changes.

Path Planning : In a video feed, to update the goal point in every subsequent frame, the idea is to “compute” the goal point based on semantic landmarks. Assume that the command robot receives is to “Go next to the shoe”. The goal-point computing pipeline will then dynamically compute the goal point based on the semantic information in the image each frame so that the robot more or less gets a consistent direction to plan to.

Grade A deliverables

Come up with the best possible method to fuse the information provided by the VLLM (traversability scores) and depthAnything (terrain score). Integrate all the different pipelines to together and make it work!

Distribution of work:

Abhinav - SAM, VLM querying

Rajesh - DepthAnything, slope estimation

Arka - VLM querying, path planning

Targets over the week:

Week Number	Tasks
1	Separate components working: DepthAnything and slope estimate from a static image, SAM segmentation and querying a VLM, A* path planning in an abstract cost map in image space.
2	Integrating the components together, debugging, video capture, have more information on how to handle the consistency of components with time
3	Final testing and presentation

Risk Assessment:

One risk could be deciding how much weight to give to VLM and slope scores each; for example, VLM may say the outdoor ramp is traversable, but the slope may be too steep, so we need to find a good way to fuse them effectively.

Something to be kept in mind is to also take into consideration the consistency of the traversability score assigned by the VLLM across frames in a video, to make sure that the traversability distribution of the “world” when we estimate it reflects this.

References:

[1] ZeST an LLM-based Zero-Shot Traversability Navigation for Unknown Environments